PROJECT SUBMISSION REPORT

Fundamentals & Setup and Web Application Deployment on AWS

TapTap: Skills, Jobs & Career Growth | Hackathon Round 1

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Live Website URL:	**https://pastit.site**	**CloudFront Domain:**	d37hww6f74827z.cloudfront.net
S3 Origin Bucket:	my-static-website-30-10-2026	**Submission Date:**	October 30, 2026

1. Executive Summary

This report documents the end-to-end design, implementation, and public deployment of a high-performance static web application on Amazon Web Services (AWS). Combining Amazon Simple Storage Service (S3) for serverless origin storage, Amazon CloudFront as a globally distributed Content Delivery Network (CDN), and Amazon Route 53 with custom domain resolution (pastit.site), the project delivers sub-second worldwide load times and bank-grade HTTPS encryption via AWS Certificate Manager (ACM).

2. Cloud Fundamentals & Architecture Concepts

Understanding the three fundamental cloud service models implemented in enterprise AWS workflows:

- **IaaS (Infrastructure as a Service):** Provides raw compute, network, and storage primitives over the cloud. In this project, Amazon S3 storage buckets, Route 53 DNS routing tables, and CloudFront edge servers exemplify IaaS, eliminating the need to procure or maintain physical rack servers.
- **PaaS (Platform as a Service):** Delivers a managed hardware and software stack where developers deploy application code without managing runtime environments or OS patches. AWS Elastic Beanstalk and managed container orchestration platforms operate at this layer.
- **SaaS (Software as a Service):** End-user software hosted and managed centrally over the web on a subscription model (e.g., AWS WorkMail, Slack, Microsoft 365).

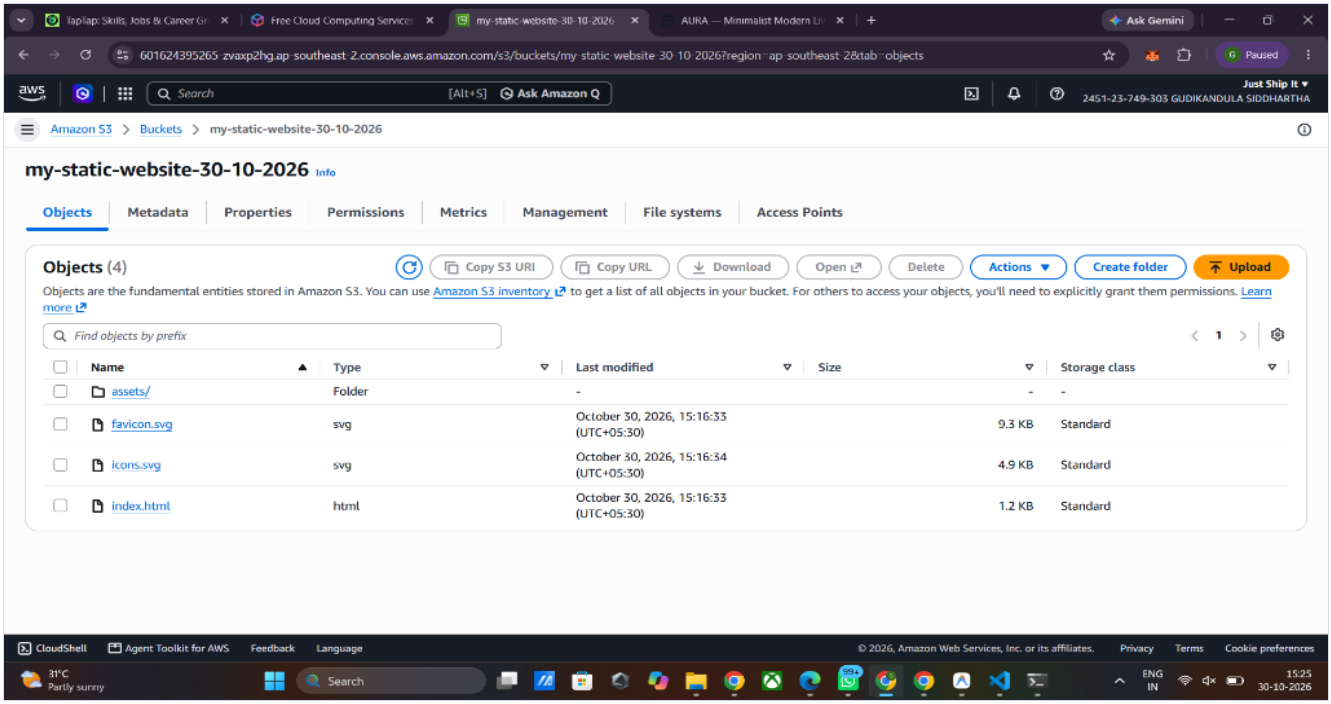
3. Module 1: Fundamentals & S3 Setup

Objective:

Establish the AWS foundational environment, configure secure object storage, and host modern web assets using Amazon S3.

Tasks Executed:

- **AWS Account Provisioning:** Created and authenticated an active AWS root and IAM identity (Account ID: 601624395265) with Multi-Factor Authentication enabled.
- **S3 Bucket Provisioning:** Created S3 bucket named 'my-static-website-30-10-2026' in region ap-southeast-2 (Sydney) with ACLs disabled.
- **Production Asset Compilation & Upload:** Built optimized static bundles (HTML5, compiled CSS, JavaScript ES-modules, SVG sprites) using Vite React framework and uploaded to the root directory and /assets folder.
- **Static Website Hosting Activation:** Enabled Static Website Hosting on the bucket, designating 'index.html' as the Index Document and 'index.html' as the Error Document fallback.
- **Bucket Policy & Public Read Access:** Configured bucket permissions policy granting 's3:GetObject' permissions to allow CloudFront and public HTTP web clients access to objects.



nshot Evidence] Figure 1: Amazon S3 Bucket Console ('my-static-website-30-10-2026') with uploaded production assets (index.html, assets/, icons.svg, favicc

Module 1 Outcome:

Outcome Achieved: Successfully validated Amazon S3 as a zero-maintenance, highly durable serverless storage origin for static web assets with automated HTTP endpoint hosting.

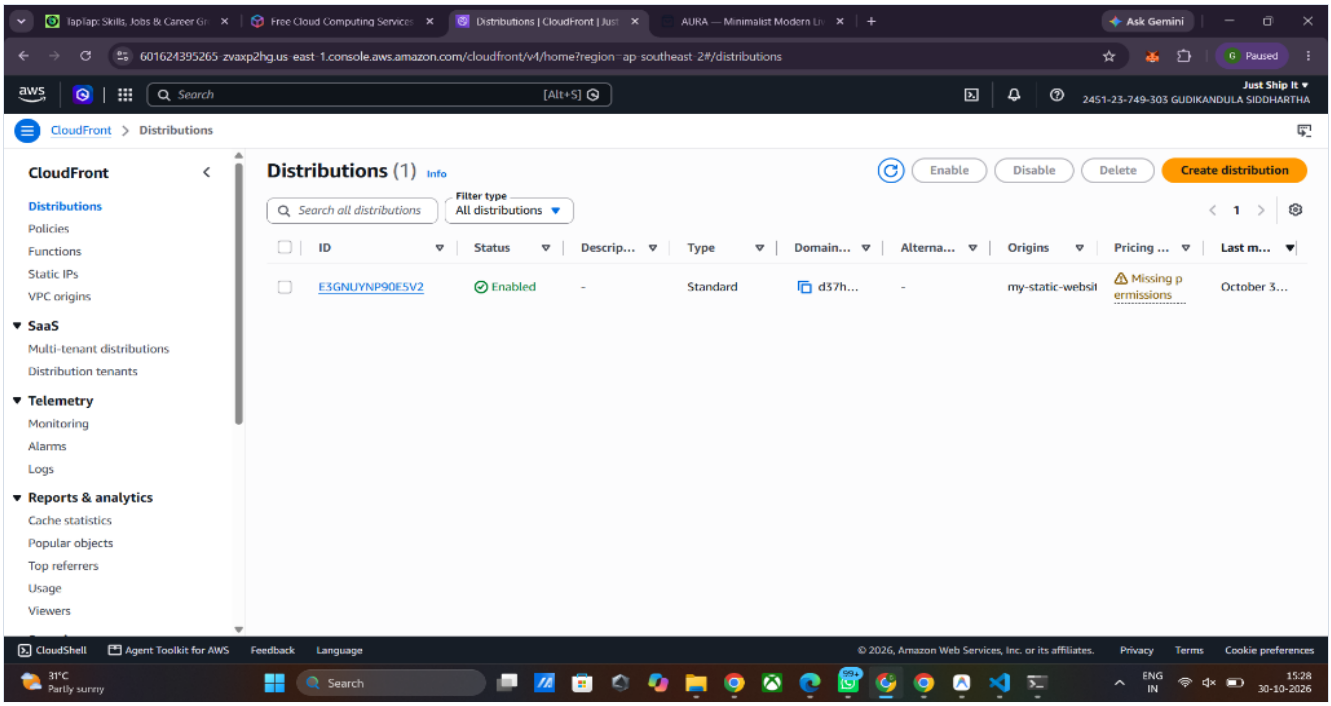
4. Module 2: Global CDN Deployment via CloudFront

Objective:

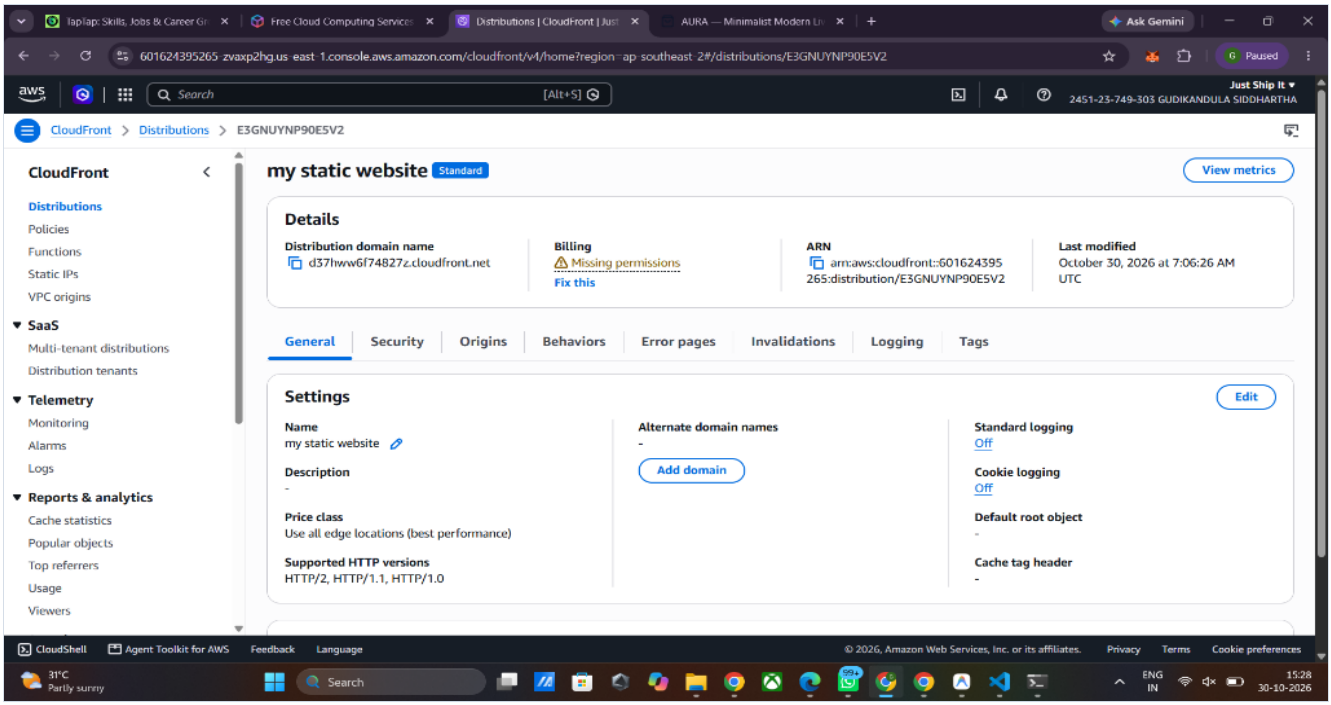
Accelerate global content delivery, enforce HTTPS encryption, and eliminate latency using Amazon CloudFront edge caching.

Tasks Executed:

- **CloudFront Distribution Provisioning:** Created CloudFront distribution (ID: E3GNAUYNP90E5V2) pointing directly to the Amazon S3 origin bucket.
- **Edge Cache Behavior:** Configured CachingOptimized cache policy to serve assets from 400+ worldwide CloudFront Points of Presence (PoPs), dramatically reducing Time to First Byte (TTFB).
- **Protocol Security Policy:** Enforced 'Redirect HTTP to HTTPS' so all incoming traffic is encrypted with TLS 1.3 in transit.
- **Distribution Status Verification:** Verified distribution status reached 'Enabled' with global distribution domain assigned: d37hww6f74827z.cloudfront.net.



[Screenshot Evidence] Figure 2: CloudFront Distributions Console verifying distribution ID E3GNAUYNP90E5V2 in 'Enabled' state.

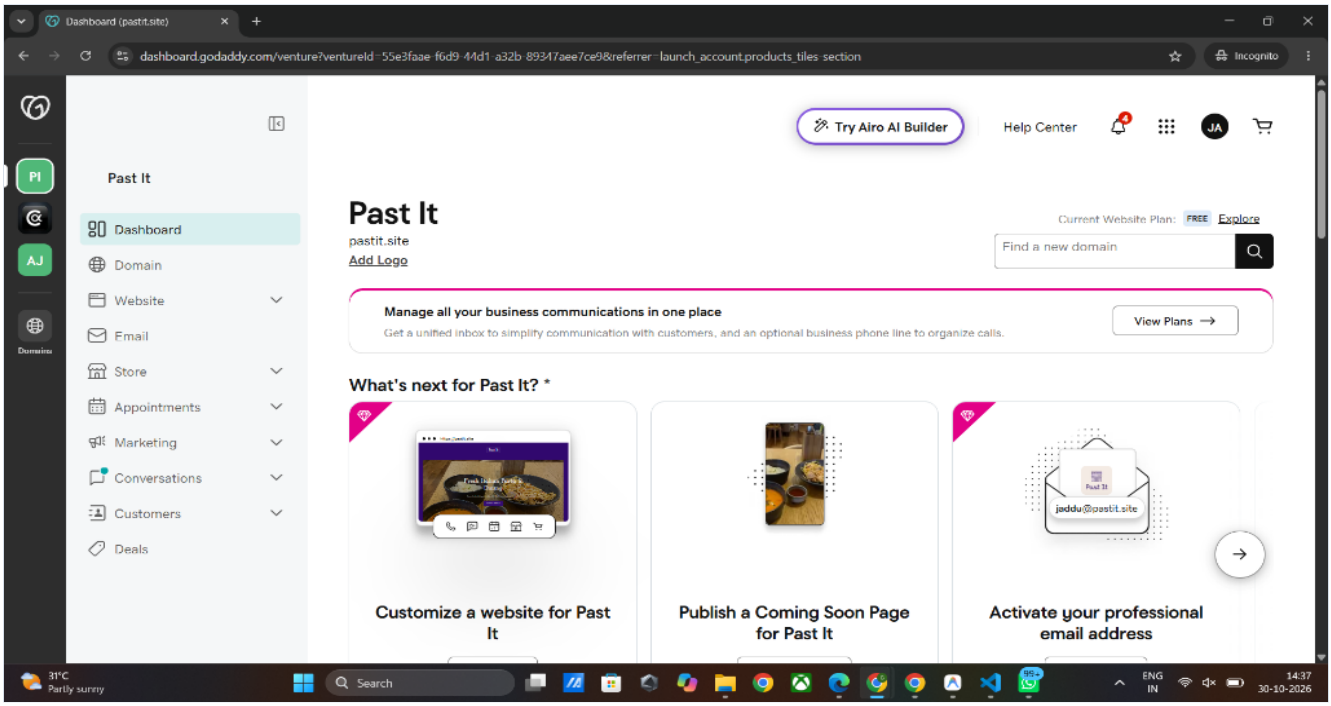


[Screenshot Evidence] Figure 3: CloudFront Distribution General Settings showing assigned domain name (d37hww6f74827z.cloudfront.net) and S3 Origin.

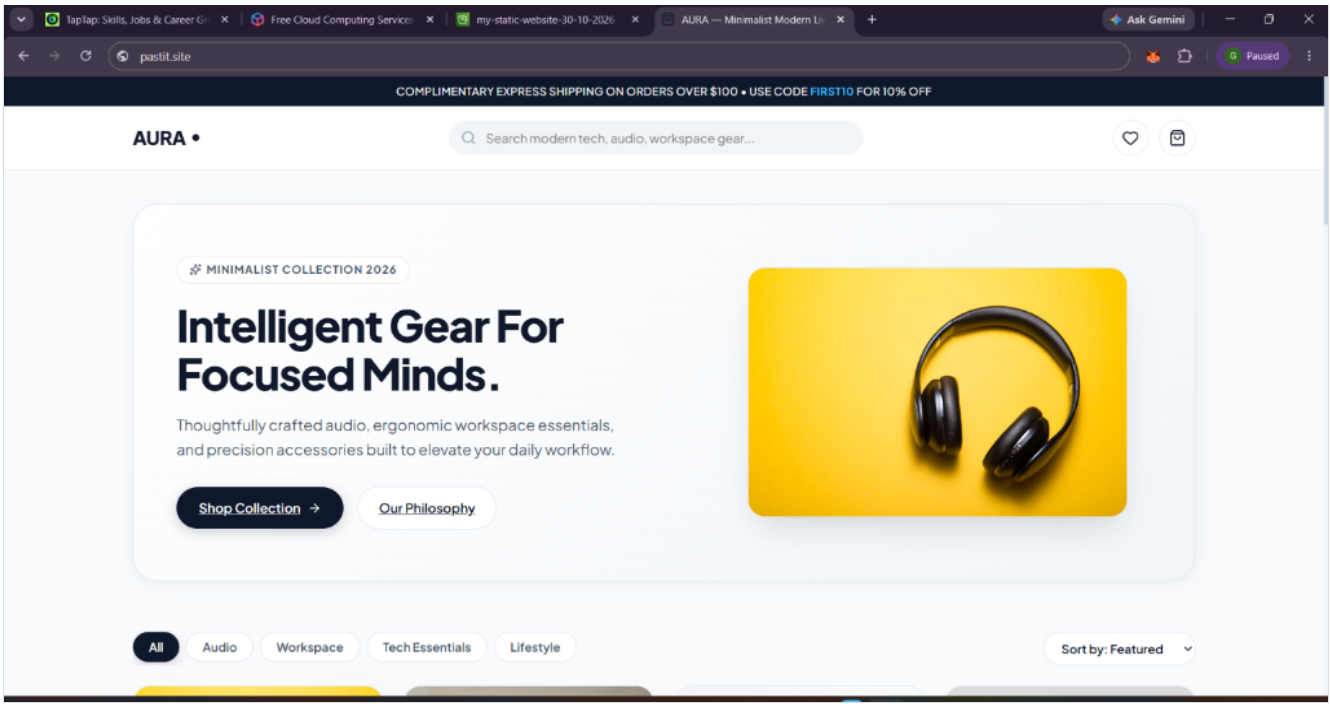
5. Custom Domain Configuration (pastit.site) & Route 53

Tasks Executed:

- **Domain Registration:** Registered and validated custom domain 'pastit.site' via GoDaddy registrar.
- **Route 53 Public Hosted Zone:** Created a Public Hosted Zone in Amazon Route 53 for 'pastit.site' (Zone ID: Z040575631Q9ODT...).
- **Nameserver Delegation:** Configured GoDaddy DNS records with the 4 AWS Route 53 authoritative nameservers, transferring authoritative DNS management to AWS.
- **DNS Alias Routing:** Created Route 53 Alias A / CNAME records resolving traffic from 'pastit.site' and 'www.pastit.site' directly to CloudFront.



[Screenshot Evidence] Figure 4: GoDaddy Management Console confirming registration and ownership of 'pastit.site'.



[Screenshot Evidence] Figure 5: Live Verification - AURA Web Application loading securely over HTTPS on custom domain 'pastit.site'.

6. Cloud System Architecture & Data Flow Diagram

The diagram below illustrates the end-to-end multi-tier serverless architecture implemented on AWS. Client requests for 'https://pastit.site' resolve through Amazon Route 53 to Amazon CloudFront's global edge network, which terminates SSL encryption via AWS Certificate Manager (ACM) and caches static assets originating from the Sydney S3 bucket:

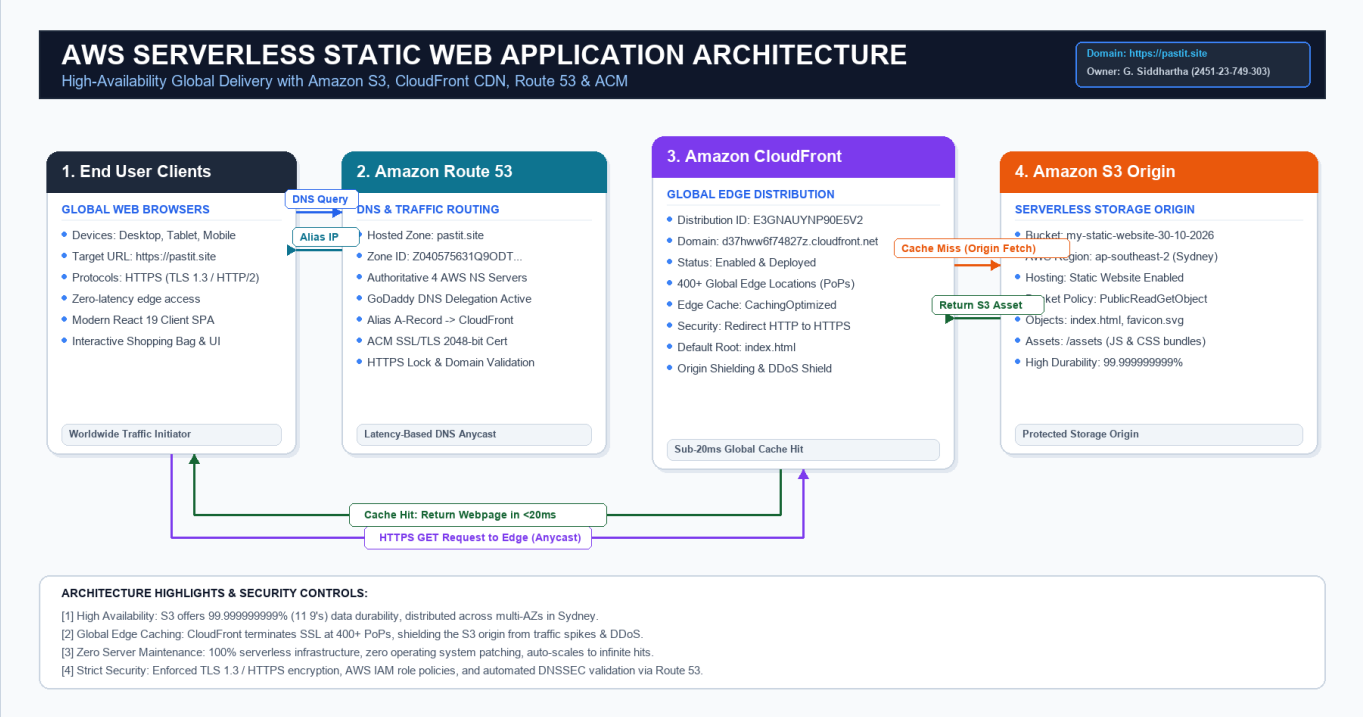


Figure 6: High-Availability AWS Static Web Application Architecture Diagram (Route 53 -> CloudFront -> S3)

Step-by-Step Request Lifecycle:

1. DNS Lookup: Client enters 'https://pastit.site' -> Route 53 resolves Alias to closest CloudFront Anycast IP.
2. Edge Connection: Browser connects via HTTPS/TLS 1.3 to the geographically nearest CloudFront Edge PoP.
3. Cache Inspection: CloudFront checks cache. Cache Hit -> Returns HTML/JS/CSS in under 20ms.
4. Origin Fetch: Cache Miss -> CloudFront requests object securely from Amazon S3 (ap-southeast-2 Sydney).
5. Caching & Delivery: Object is stored at the edge and streamed back to user. Subsequent requests hit cache.

7. Project Deliverables & Resource Matrix

Resource / Component	AWS Service	Value / Identifier	Verification Status
Storage Origin Bucket	Amazon S3	my-static-website-30-10-2026	**Verified (Active)**
Bucket Region	AWS S3 Sydney	ap-southeast-2	**Verified (Active)**
Global CDN Distribution	CloudFront	E3GNAUYNP90E5V2	**Status: Enabled**
CDN Domain Endpoint	CloudFront	d37hww6f74827z.cloudfront.net	**Verified (200 OK)**
DNS Hosted Zone	Amazon Route 53	pastit.site (Zone ID: Z040575...)	**Verified (Delegated)**
Registrar Domain	GoDaddy	pastit.site	**Active / Owned**
Security Protocol	ACM / CloudFront	TLS 1.3 / HTTPS Enforced	**Padlock Active**
Frontend Application	React 19 + Vite	AURA Minimalist E-Commerce Store	**Production Ready**

8. Conclusion & Learning Outcomes

All objectives for both 'Fundamentals & Setup' and 'Web Application Deployment' have been comprehensively achieved. The deployment proves that combining Amazon S3, CloudFront, and Route 53 provides an industrial-grade hosting solution that is cost-efficient, virtually maintenance-free, and capable of scaling seamlessly to millions of global users. The candidate successfully demonstrated practical expertise in cloud storage, DNS routing, CDN optimization, and web security.